Online Appendix: Multi-Strategy GraphRAG for Financial Institutions: Security, Grounding, and Token Economy

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Abstract

This document provides supplementary material for the paper "Multi-Strategy GraphRAG for Financial Institutions: Security, Grounding, and Token Economy," including the complete financial domain ontology specification, triple extraction prompt templates, the full set of benchmark questions, additional experimental results, and implementation documentation.

1. Financial domain ontology specification

This appendix provides the complete specification of the financial domain ontology used in the proposed GraphRAG system.

1.1. Entity types

Table **??** provides the complete list of entity types with descriptions, examples, and the number of instances in the evaluation corpus.

1.2. Relationship types

Table **??** provides the complete list of relationship types with source/target constraints and cardinality.

1.3. Entity alias dictionary

The entity resolution module uses an alias dictionary with approximately 130 entries to handle common abbreviations and alternative names. Table **??** provides a representative sample.

2. Triple extraction prompt templates

2.1. System prompt

The following system prompt is used for all triple extraction calls:

Listing 1: System prompt for triple extraction.

```
You are a financial knowledge graph extraction engine.
Your task is to extract structured entity-relationship
triples from financial and regulatory text.

STRICT RULES:
1. Use ONLY the entity types and relationship types
   provided in the schema below.
2. Entity names must be in PascalCase.
3. Assign a confidence score between 0.0 and 1.0 to
   each triple based on how explicitly the relationship
   is stated in the text.
4. Include textual evidence: a direct quote from the
   source text that supports each triple.
5. Do not infer relationships not explicitly stated or
   strongly implied in the text.

ENTITY TYPES:
- Regulation: Legal or regulatory framework
- Article: Specific provision within a regulation
- Requirement: Obligation, standard, or criterion
- RiskCategory: Classification of risk type
- FinancialMetric: Quantitative financial measure
- Company: Corporate entity
- Subsidiary: Corporate subsidiary or division
- Jurisdiction: Geographic or legal jurisdiction
- Control: Internal control mechanism
- AuditFinding: Result of audit assessment

RELATIONSHIP TYPES:
- REGULATES: Regulation governs entity/jurisdiction
- REQUIRES: Regulation/article mandates requirement
- DEFINES: Article establishes requirement/metric
- EXPOSES_TO: Entity has exposure to risk
```

```
- SUBSIDIARY_OF: Subsidiary belongs to company
- OPERATES_IN: Entity operates in jurisdiction
- AUDITED_BY: Entity has associated audit finding
- MITIGATES: Control addresses risk/finding
- REFERENCES: Document cross-references another
- SUPERSEDES: Regulation replaces another
- COMPOSED_OF: Regulation contains articles
- REPORTS_TO: Subsidiary reports to company
- VALIDATED_BY: Requirement verified by control
- COMPLIES_WITH: Company meets requirement/article
```

2.2. User prompt template

Listing 2: User prompt template for triple extraction.

```
Extract entity-relationship triples from the following
text chunk.

Chunk type: {chunk_type}
Source file: {source_file}
---
{chunk_content}
---

Return a JSON object with an array named "triples".
Each triple must have these fields:
- subject (string)
- subject_type (string, from entity types)
- predicate (string, from relationship types)
- object (string)
- object_type (string, from entity types)
- confidence (float, 0.0 to 1.0)
- evidence (string, quote from text)
```

3. Financial benchmark questions

Table **??** presents the complete set of 50 benchmark questions used in Experiment 5, organized by category with difficulty level and required knowledge

graph traversal depth.

Table 4: Complete 50-question financial benchmark with difficulty and traversal depth.

#	Category	Question	Diff.	Hops
1	Compl.	Does Basel III Pillar 1 require a minimum CET1 ratio of 4.5%?	S	1
2	Compl.	Which LGPD articles govern processing of financial credit data?	M	2
3	Compl.	What are the GDPR Art. 25 privacy-by-design requirements?	S	1
4	Compl.	How does LGPD Art. 46 define security measures for data processing?	M	2
5	Compl.	Which Basel III provisions address countercyclical capital buffers?	M	2
6	Compl.	What SOX Section 404 requirements apply to internal controls?	S	1
7	Compl.	How do LGPD and GDPR differ on data subject access rights?	C	2
8	Compl.	What are the Basel III Pillar 2 supervisory review requirements?	M	2
9	Compl.	Which LGPD articles require DPO appointment for financial institutions?	M	2
10	Compl.	What Basel III disclosure requirements apply under Pillar 3?	M	2
11	Compl.	How does GDPR Art. 35 DPIA apply to financial profiling?	C	2
12	Compl.	What are the capital conservation buffer requirements?	S	1

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#	Category	Question	Diff.	Hops
13	Compl.	Which LGPD provisions govern international data transfers?	M	2
14	Compl.	How do Basel III and CRD VI interact for EU-operating banks?	C	3
15	Compl.	What are the GDPR requirements for breach notification?	S	1
16	Fin.	What was Company X's total revenue in FY2024?	S	1
17	Fin.	Compare net income margins of Company A vs Company B.	M	2
18	Fin.	What are the main risk factors reported by Company X?	S	1
19	Fin.	How did Company X's debt-to-equity ratio change YoY?	M	2
20	Fin.	Which companies report the highest R&D spending?	M	2
21	Fin.	What is the operating cash flow trend for Company X?	M	2
22	Fin.	Compare revenue growth across the technology sector.	M	2
23	Fin.	What goodwill impairments were reported in FY2024?	S	1
24	Fin.	Which companies have the lowest capital adequacy ratio?	M	2
25	Fin.	What are Company X's key subsidiaries and their revenues?	M	2
26	Fin.	Compare operating margins across financial sector companies.	M	2

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#	Category	Question	Diff.	Hops
27	Fin.	What contingent liabilities does Company X disclose?	S	1
28	Multi	Which subsidiaries of Company X have credit risk exposure under Basel III?	C	3
29	Multi	Trace the regulatory chain from LGPD Art. 46 to specific controls.	C	3
30	Multi	Which EU subsidiaries of Brazilian banks must comply with CRD VI?	C	3
31	Multi	What controls mitigate the risks identified in audit finding F-103?	C	3
32	Multi	Which regulations apply to cross-border data transfers by Company X?	C	3
33	Multi	What are the capital requirements for Company X's EU operations?	C	3
34	Multi	Which audit findings relate to LGPD compliance gaps?	C	3
35	Multi	Trace the subsidiaries exposed to operational risk in emerging markets.	C	3
36	Multi	What controls validate Basel III Pillar 1 requirements for Company X?	C	3
37	Multi	Which jurisdictions create dual regulatory obligations for Company X?	C	3
38	Comp.	Compare LGPD and GDPR data minimization requirements.	M	2
39	Comp.	How do Basel III and Basel II differ on operational risk capital?	C	3
40	Comp.	Compare SOX and LGPD audit trail requirements.	M	2

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#	Category	Question	Diff.	Hops
41	Comp.	What are the differences in data breach notification between LGPD and GDPR?	M	2
42	Comp.	Compare capital adequacy requirements across three jurisdictions.	C	3
43	Comp.	How do internal vs. external audit requirements differ under SOX?	M	2
44	Comp.	Compare Company A and Company B risk factor disclosures.	M	2
45	Comp.	What are the differences between LGPD and GDPR DPO requirements?	M	2
46	Audit	List all high-risk findings from the Q4 2024 operational risk audit.	S	1
47	Audit	Which controls mitigate the risks in finding F-103?	M	2
48	Audit	What is the remediation status of findings from the 2024 audit cycle?	M	2
49	Audit	Which audit findings have been open for more than 90 days?	M	2
50	Audit	What is the overall risk assessment for the payments control environment?	C	3

Legend: S = Simple, M = Moderate, C = Complex. Hops = minimum knowledge graph traversal depth required for a complete answer.

4. Additional experimental results

4.1. Detailed hallucination results by category

Table **??** presents the complete breakdown of faithfulness, factual accuracy, and grounding rate by question category and system.

4.2. Latency percentile distribution

Table **??** presents the full percentile distribution of latency measurements for each system configuration.

4.3. Red-team attack breakdown

Table **??** provides the detailed breakdown of red-team security evaluation results by attack sub-category.

5. Implementation guide and API documentation

5.1. System requirements

5.2. API endpoints

Table **??** documents the primary API endpoints of the system.

5.3. Configuration

The system is configured through environment variables and a YAML configuration file. Key configuration parameters include:

Listing 3: Key configuration parameters.

```
# Neo4j connection
NEO4J_URI=bolt://localhost:7687
NEO4J_USER=neo4j
NEO4J_PASSWORD=***

# PostgreSQL connection
DATABASE_URL=postgresql+asyncpg://user:pass@localhost/graphrag

# Claude API
ANTHROPIC_API_KEY=***
LLM_MODEL=claude-haiku-4-5-20251001
USE_BATCH_API=true

# Retrieval settings
MAX_CONTEXT_TOKENS=4000
EMBEDDING_MODEL=all-MiniLM-L6-v2
```

```
FUZZY_THRESHOLD=0.85
COMMUNITY_RESOLUTION=1.0

# Security
JWT_SECRET=***
JWT_ALGORITHM=HS256
JWT_EXPIRATION=3600
```

5.4. Docker deployment

The system can be deployed using Docker Compose for development and testing:

Listing 4: Docker Compose configuration for local deployment.

```
# docker-compose.yml
services:
  neo4j:
    image: neo4j:5-community
    ports:
      - "7474:7474"
      - "7687:7687"
    environment:
      NEO4J_AUTH: neo4j/password
      NEO4J_PLUGINS: '["apoc"]'

  postgres:
    image: postgres:16
    ports:
      - "5432:5432"
    environment:
      POSTGRES_DB: graphrag
      POSTGRES_USER: graphrag
      POSTGRES_PASSWORD: password

  api:
    build: .
    ports:
      - "8000:8000"
```

```
    depends_on:
      - neo4j
      - postgres
    environment:
      NEO4J_URI: bolt://neo4j:7687
      DATABASE_URL: postgresql+asyncpg://graphrag:password@postgres
          /graphrag
```

Table 1: Complete financial domain ontology: entity types with descriptions and instance counts.

Entity Type	**Description**	**Examples**	**Count**
Regulation	Legal/regulatory framework	Basel III, LGPD, GDPR, SOX, CRD VI	15
Article	Specific provision within a regulation	LGPD Art. 18, GDPR Art. 25, SOX Sec. 404	120
Requirement	Obligation or standard	Capital adequacy ratio, Data minimization	85
RiskCategory	Classification of risk	Credit risk, Operational risk, Market risk	12
FinancialMetric	Quantitative measure	Revenue, Net income, CET1 ratio	45
Company	Corporate entity	S&P 100 companies, regulated banks	50
Subsidiary	Corporate subsidiary	Technology divisions, regional subsidiaries	80
Jurisdiction	Geographic/legal jurisdiction	Brazil, European Union, United States	20
Control	Internal control mechanism	Segregation of duties, Access control policy	100
AuditFinding	Result of audit assessment	High-risk findings, observations	200

Table 2: Complete financial domain ontology: relationship types with constraints.

Relationship	**Source Types**	**Target Types**	**Card.**
Regulates	Regulation	Company, Jurisdiction	M:N
Requires	Regulation, Article	Requirement	1:N
Defines	Article	Requirement, FinancialMetric	1:N
ExposesTo	Company, Subsidiary	RiskCategory	M:N
SubsidiaryOf	Subsidiary	Company	N:1
OperatesIn	Company, Subsidiary	Jurisdiction	M:N
AuditedBy	Company, Subsidiary	AuditFinding	1:N
Mitigates	Control	RiskCategory, AuditFinding	M:N
References	Article, Regulation	Article, Regulation	M:N
Supersedes	Regulation	Regulation	1:1
ComposedOf	Regulation	Article	1:N
ReportsTo	Subsidiary	Company	N:1
ValidatedBy	Requirement	Control	M:N
CompliesWith	Company	Requirement, Article	M:N

Table 3: Sample aliases from the entity resolution dictionary.

Canonical Name	**Type**	**Aliases**
Basel III	Regulation	Basel 3, Basel III Framework, B3
LGPD	Regulation	Lei Geral de Protecao de Dados, Lei 13709
GDPR	Regulation	General Data Protection Regulation, RGPD
SOX	Regulation	Sarbanes-Oxley, Sarbanes Oxley Act
Credit Risk	RiskCategory	Risco de credito, Default risk
CET1 Ratio	FinancialMetric	Common Equity Tier 1, CET1

Table 5: Detailed hallucination metrics by question category (200-question regulatory dataset).

Category	**System**	**Faith.**	**Acc.**	**Ground.**
Compliance (n=80)	Ours	87.54%	84.24%	96.61%
	HybridRAG	82.46%	78.11%	72.18%
	MS GraphRAG	80.57%	76.34%	70.52%
	VectorRAG	72.78%	68.74%	55.17%
	No Retrieval	46.54%	41.23%	0.00%
Fin. Analysis (n=50)	Ours	87.97%	84.52%	96.48%
	HybridRAG	83.09%	79.04%	72.26%
	MS GraphRAG	79.59%	74.87%	69.74%
	VectorRAG	72.92%	68.82%	55.12%
	No Retrieval	46.44%	41.08%	0.00%
Multi-hop (n=40)	Ours	86.24%	82.70%	96.56%
	HybridRAG	81.35%	77.45%	72.74%
	MS GraphRAG	79.22%	74.38%	70.42%
	VectorRAG	73.40%	68.56%	55.55%
	No Retrieval	45.78%	40.54%	0.00%
Comparison (n=20)	Ours	87.10%	83.44%	95.85%
	HybridRAG	83.23%	79.52%	71.90%
	MS GraphRAG	78.66%	74.12%	69.45%
	VectorRAG	72.96%	68.60%	54.80%
	No Retrieval	45.90%	40.62%	0.00%
Audit (n=10)	Ours	87.60%	83.88%	96.80%
	HybridRAG	82.01%	77.82%	71.46%
	MS GraphRAG	79.46%	74.62%	70.00%
	VectorRAG	71.14%	67.24%	53.80%
	No Retrieval	50.20%	44.56%	0.00%

Table 6: Latency percentile distribution (milliseconds) across all system configurations.

Configuration	**p25**	**p50**	**p75**	**p90**	**p95**
Ours (serial)	3,654	4,200	4,892	5,445	5,868
Ours (parallel)	1,542	1,785	2,034	2,287	2,434
Ours (par.+cache)	678	797	942	1,067	1,142
VectorRAG	1,024	1,182	1,378	1,592	1,716
MS GraphRAG	1,845	2,127	2,456	2,712	2,886
HybridRAG	1,267	1,469	1,724	1,945	2,102
No Retrieval	542	653	784	892	962

Table 7: Detailed red-team attack results by sub-category.

Campaign	**Sub-category**	**Queries**	**Success**	**Rate**
Cross-tenant	Direct access	25	0	0.0%
	Graph traversal	25	0	0.0%
	Cypher injection	25	0	0.0%
	Privilege escalation	25	0	0.0%
Entity extraction	Structure discovery	15	1	6.7%
	Targeted extraction	15	1	6.7%
	Inference attacks	10	0	0.0%
	Exhaustive enumeration	10	0	0.0%
Membership inference	Existence probes	20	12	60.0%
	Semantic probes	15	8	53.3%
	Differential probes	15	8	53.3%

Table 8: System requirements for deploying the proposed GraphRAG architecture.

Component	**Version**	**Purpose**
Neo4j	5.x+	Knowledge graph storage and vector index
PostgreSQL	16+	Metadata, sessions, cache, audit logs
Python	3.10+	Runtime environment
Claude API	Haiku/Sonnet	Triple extraction and response generation
sentence-transformers	all-MiniLM-L6-v2	Embedding generation (384 dimensions)
igraph + leidenalg	0.11+ / 0.10+	Community detection
FastAPI	Latest	REST API framework

Table 9: Primary API endpoints for the GraphRAG system.

Endpoint	**Method**	**Description**
`/auth/login`	POST	Authenticate user, return JWT token
`/chat`	POST	Submit query with context, receive response
`/chat/sessions`	GET	List user's chat sessions
`/graph/query`	POST	Execute direct Cypher query
`/graph/stats`	GET	Knowledge graph statistics
`/analytics/usage`	GET	API usage and cost analytics
`/health`	GET	System health check